Code No: 136EV **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year II Semester Examinations, July - 2023
DATA STRUCTURES
(Common to EEE, CSE)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

used paper July/Aug-2023

PART – A

(25 Marks)

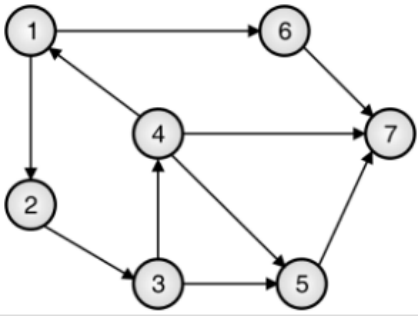
1.a)	What is the time complexity of linear loop and quadratic loop?	[2]
b)	Give the classification of data structures.	[3]
c)	What are the advantages of circular queues compared with linear queue?	[2]
d)	Convert the following infix expression into postfix expression. A+B*C/D+E-F+G	[3]
e)	Define Max Priority queue.	[2]
f)	Define tree, binary tree and heap tree.	[3]
g)	Compare linear search and binary search.	[2]
h)	Write the time complexity of selection sort, Quick sort and Radix sort.	[3]
i)	Define Red-Black Tree and Splay Tree.	[2]
j)	Construct the binary search tree using the following traversal sequences. Inorder Traversal: 10 20 30 100 150 200 300 Preorder Traversal: 100 20 10 30 200 150 300	[3]

PART – B

(50 Marks)

2.a)	Write a function to delete an element from the front of the linked list.	
b)	Write an algorithm to insert an element at the front of doubly linked list.	[5+5]
	OR	
3.a)	Write a function to print transpose of a sparse matrix.	
b)	Write a function to concatenate two singly linked lists.	[5+5]
4.a)	Write the program to implement operations of stack using linked lists.	
b)	Explain insert and delete operations of circular queue.	[5+5]
	OR	
5.a)	Write the program to implement operations of queue using arrays.	
b)	Write the program to evaluate the postfix expression.	[5+5]

6.a) Write a procedure to insert an element into binary tree.
b) Write the traversal sequences of the following Graph using BFS and DFS. [5+5]



OR

7.a) Write a function to perform delete operation in max heap.
b) Write functions to perform the following operations of binary tree:
i) pre order traversal ii) in order traversal. [5+5]

8.a) What is hashing? How hashing can used to insert and search elements into a hash table.
b) Write a program to implement selection sort. [5+5]

OR

9.a) Write a program to perform quick sort.
b) Write a program to search for an element using linear search. [5+5]

10.a) Write a function to delete an element from a binary search tree.
b) Explain various rotations of the Red-Black tree. [5+5]

OR

11.a) Write the process to insert and delete an element in AVL Tree.
b) Explain the procedure to delete an element from B-Tree. [5+5]

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